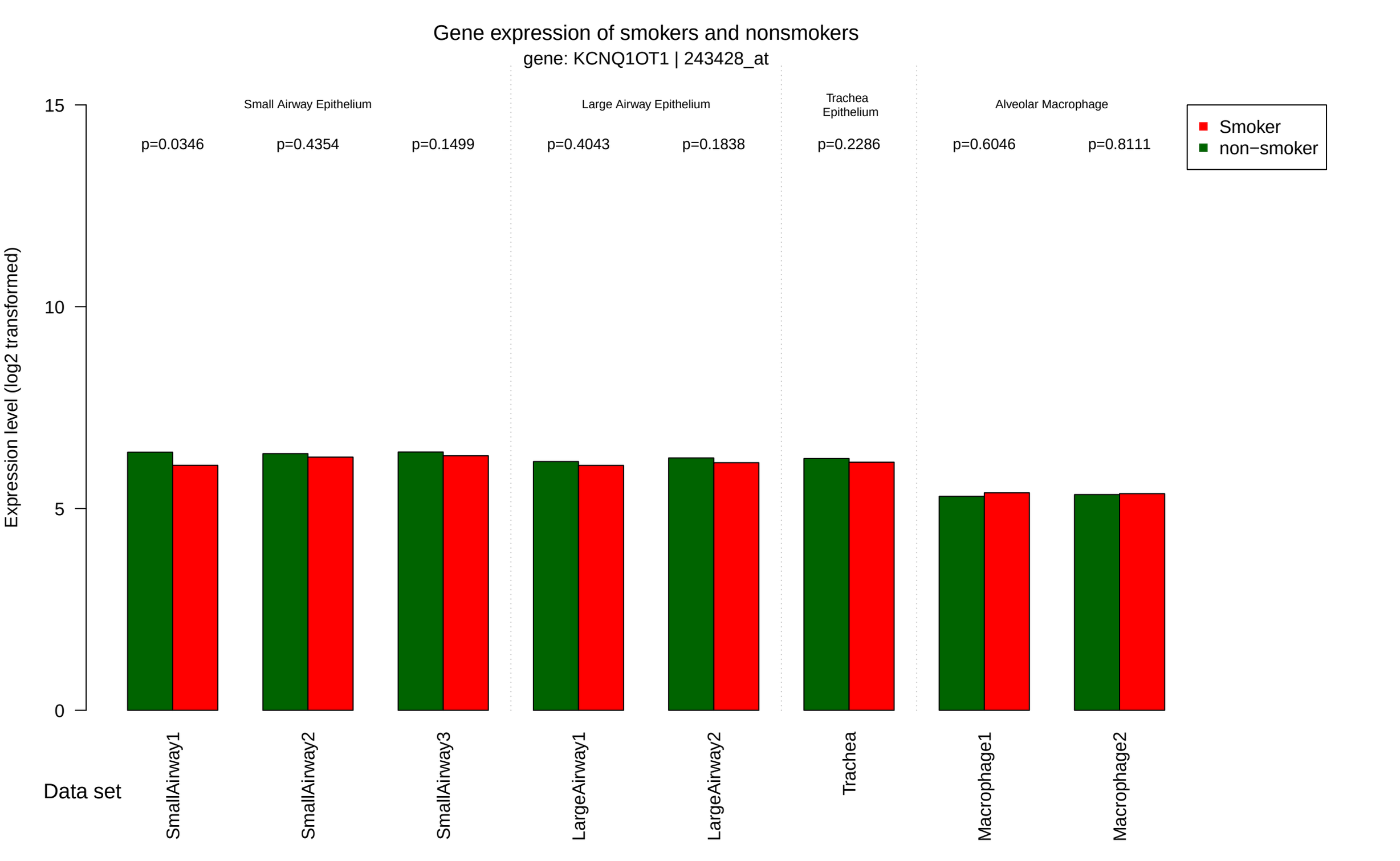
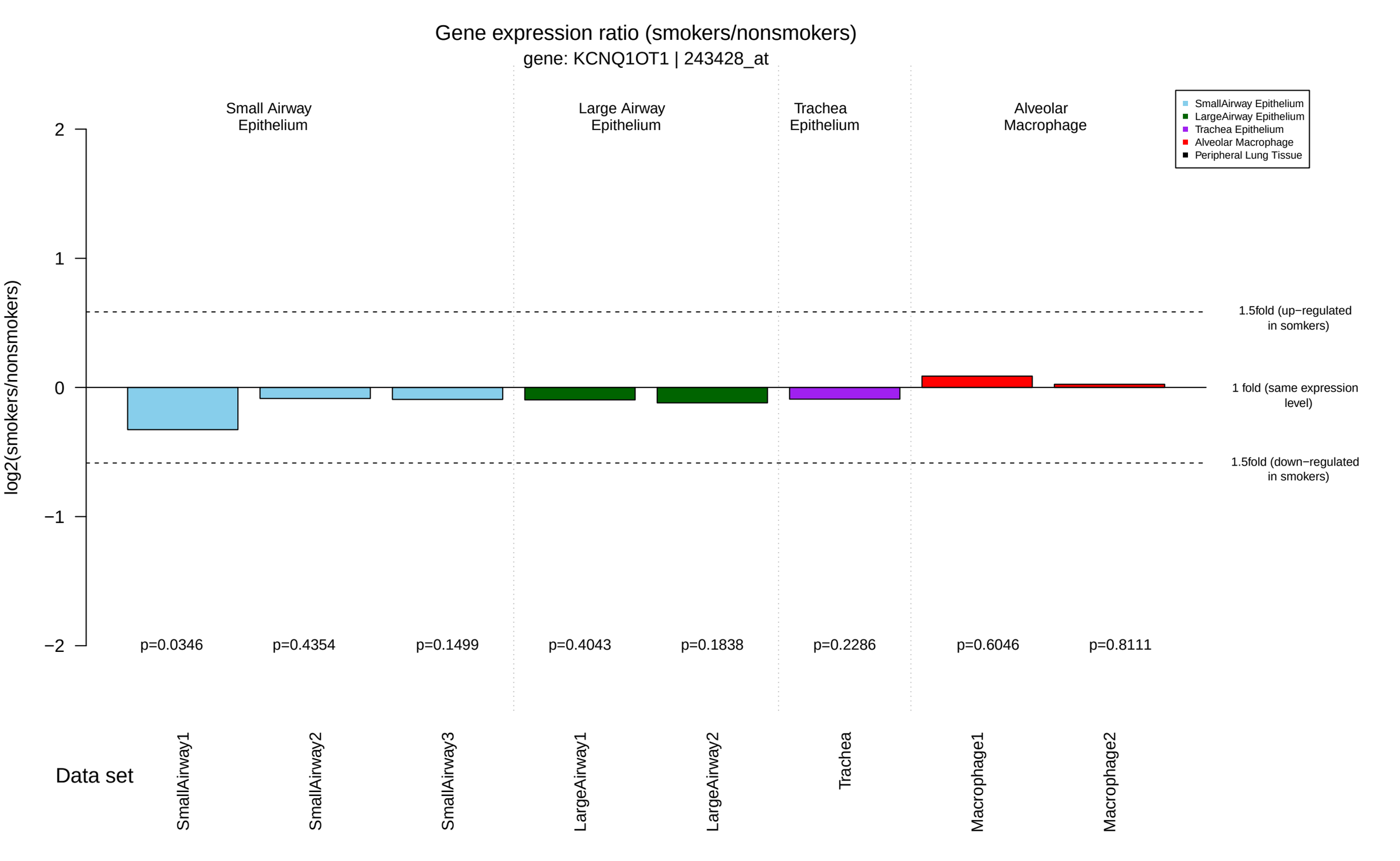
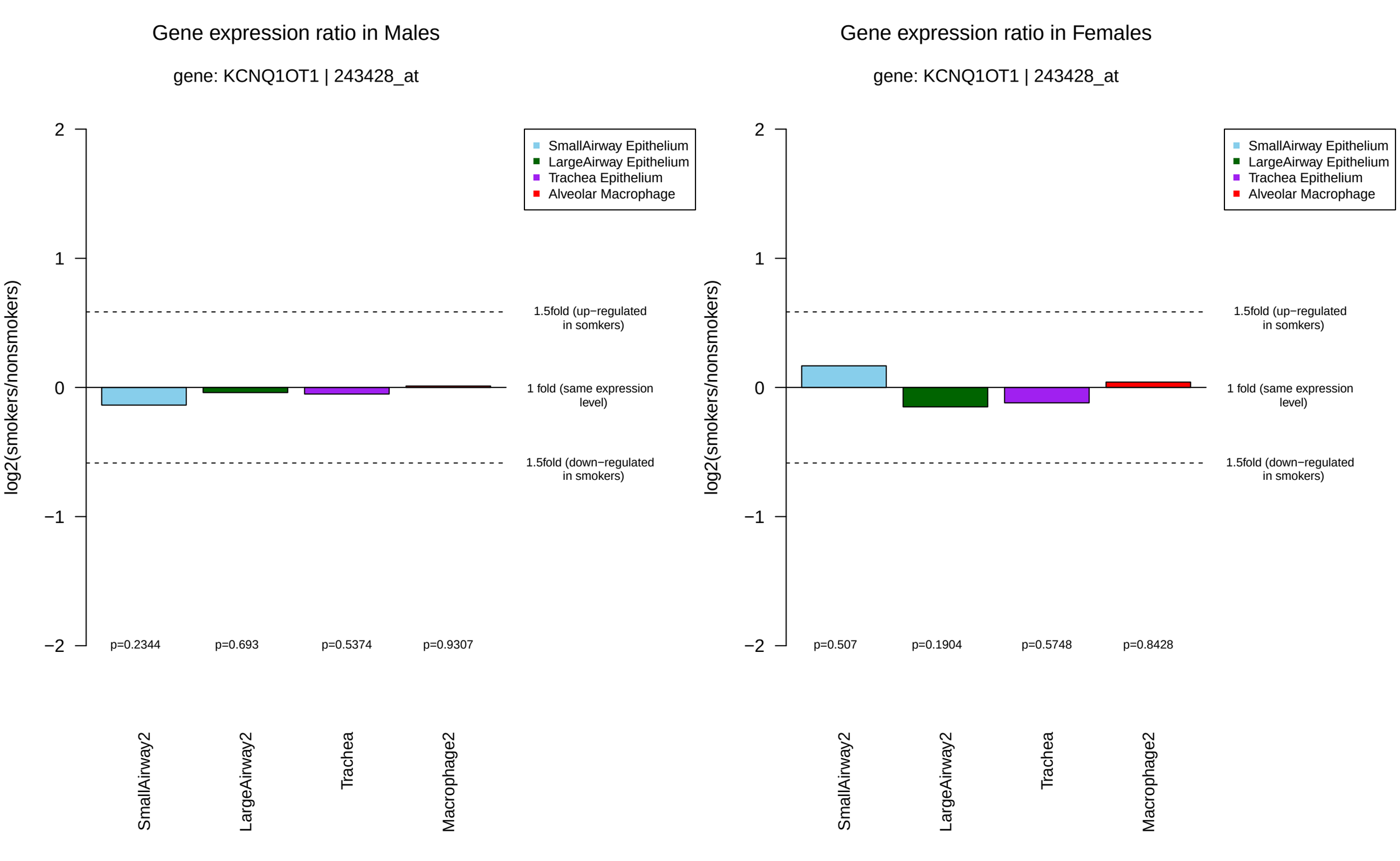
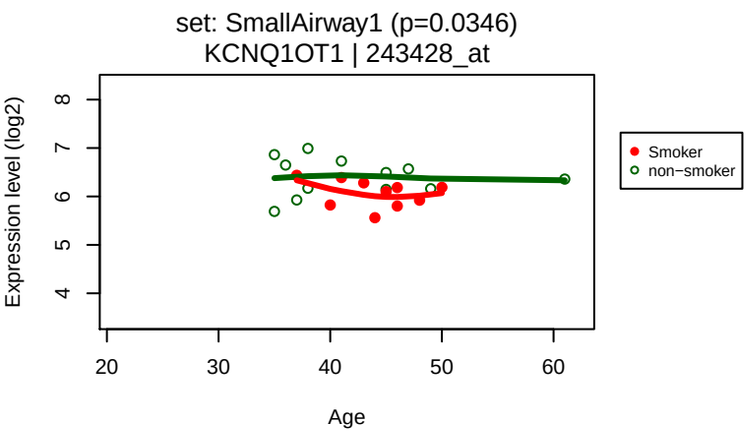
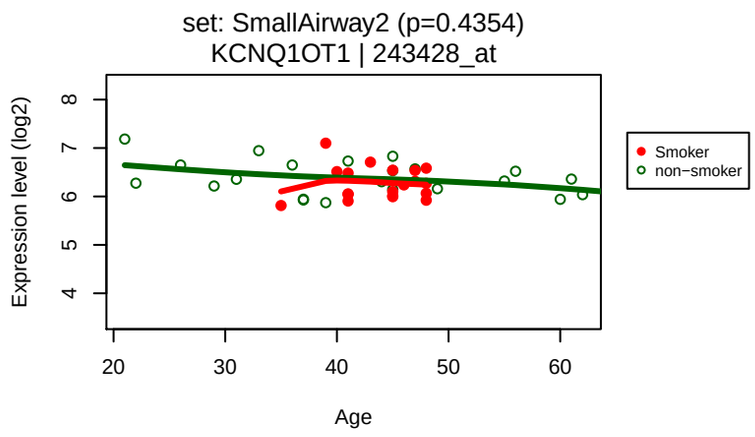
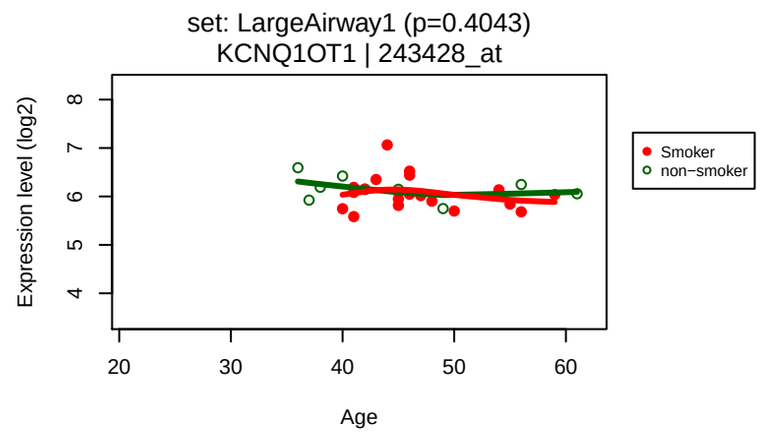

Gene expression of smokers and nonsmokers
gene: KCNQ1OT1 | 243428_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0346
p=0.4354
p=0.1499
p=0.4043
p=0.1838
p=0.2286
p=0.6046
p=0.8111
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

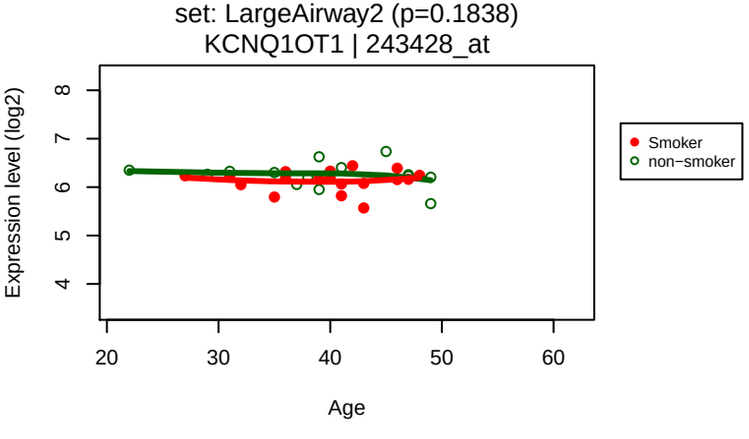

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ1OT1 | 243428_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0346
p=0.4354
p=0.1499
p=0.4043
p=0.1838
p=0.2286
p=0.6046
p=0.8111
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

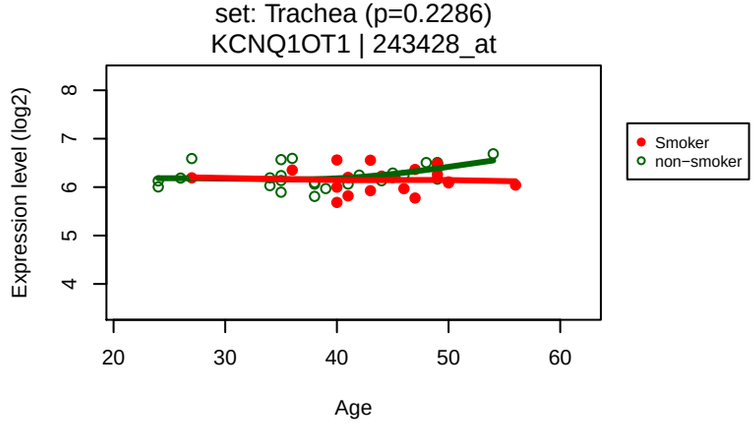

Gene expression ratio in Males
gene: KCNQ1OT1 | 243428_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2344
p=0.693
p=0.5374
p=0.9307
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ1OT1 | 243428_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.507
p=0.1904
p=0.5748
p=0.8428
SmallAirway2
LargeAirway2
Trachea
Macrophage2

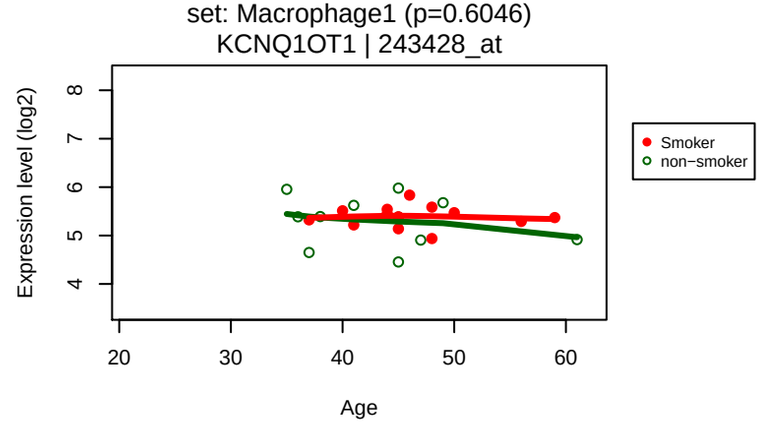

set: SmallAirway1 (p=0.0346)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker

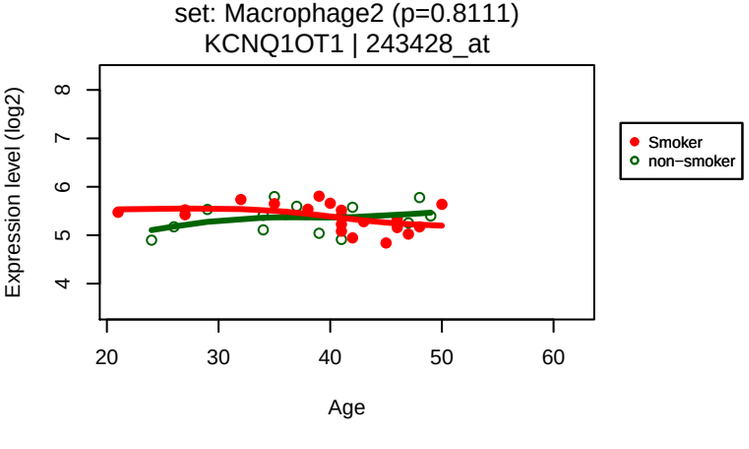

set: SmallAirway2 (p=0.4354)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker

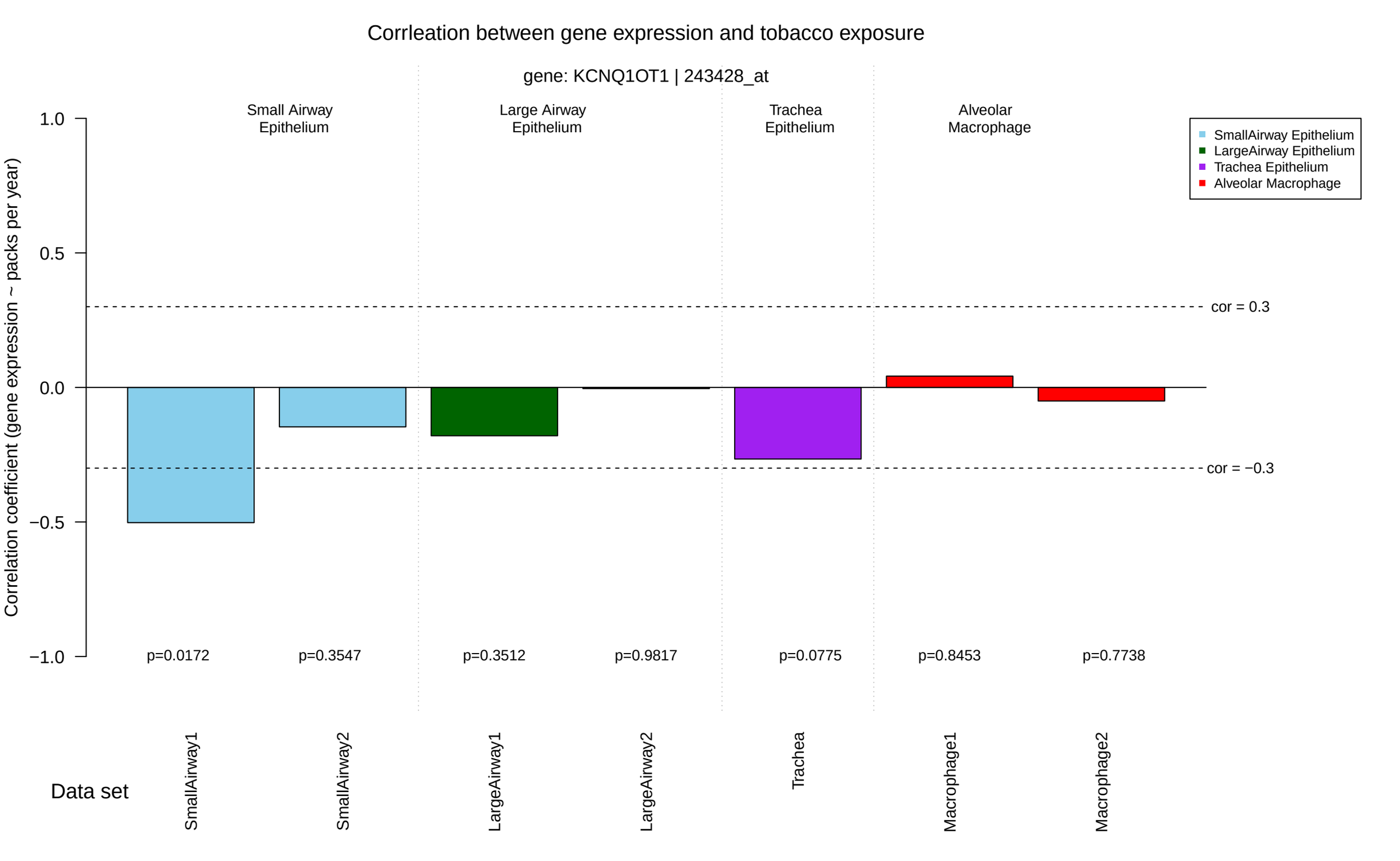

set: LargeAirway1 (p=0.4043)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1838)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2286)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6046)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8111)
KCNQ1OT1 | 243428_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ1OT1 | 243428_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0172
p=0.3547
p=0.3512
p=0.9817
p=0.0775
p=0.8453
p=0.7738
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2